



Comparison of Therapeutic Effect of Systematic Growth Factor Microneedle Treatment in the Patients with Male Pattern Hair Loss by Varying Depth of Absorption

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Background

- Reports about about growth factor treatment on androgenetic alopecia(AGA) patients:
 - YB Lee et al reported growth factor therapy(fixed 0.5mm depth) was effective and safe for treatment of alopecia areata, 1week interval for 5 weeks in 2012.
 - BI Ro et al. reported growth factor treatment on the AGA patients with microneedle(fixed 0.7mm depth) for 6 months at the 7th World Congress for Hair Research in 2013.
- Both of reports shows that SGF-57 treatment is effective and safe but there is a lack of study on the most effective depth of microneedle.
- Herein, we investigated the difference of effects according to the depth of microneedle.

Materials and Methods

- **Patients:**
 - 3months period from September, 2013 to December, 2013
 - Male Pattern Hair Loss(MPHL; 5 patients) III, IV, V and Female Pattern Hair Loss(FPHL; 6 patients) II were included.
- **Treatment:**
 - Topical application of growth factor solution into scalp by microneedle.
 - Split study : Dividing scalp by the right side and left side;
 - Right side ; 0.5mm depth of microneedle.
 - Left side ; 0.3mm depth of microneedle.
 - Growth factor (SGF-57; Aesmed Co. Ltd., Seoul, Korea)
 - bFGF , IGF -1, VEGF, stem cell factor, KGF -2, superoxide dismutase-1, and Noggin.
 - Microneedle (Dr Back 10 story FNS FN-1; Dongbang Medicare, Bundang, Korea) ; adjust depth.
 - Total 6 times (3 months) in every other week.
- **Phototrichogram(Folliscope 2.8, Lead M, Seoul, Korea)** in fixed area marked with tattoo to measure hair density and thickness at the baseline and at the end of treatment.
- **Statistics:**
 - Wilcoxon matched-pairs signed-rank test
 - Student's t-test

Results

Patient's characteristics

Androgenetic Alopecia (Total 11 patients)	Number	Min. age	Max. age
Male	6 patients	26	49
Female	5 patients	34	49

Effect of SGF treatment with microneedling

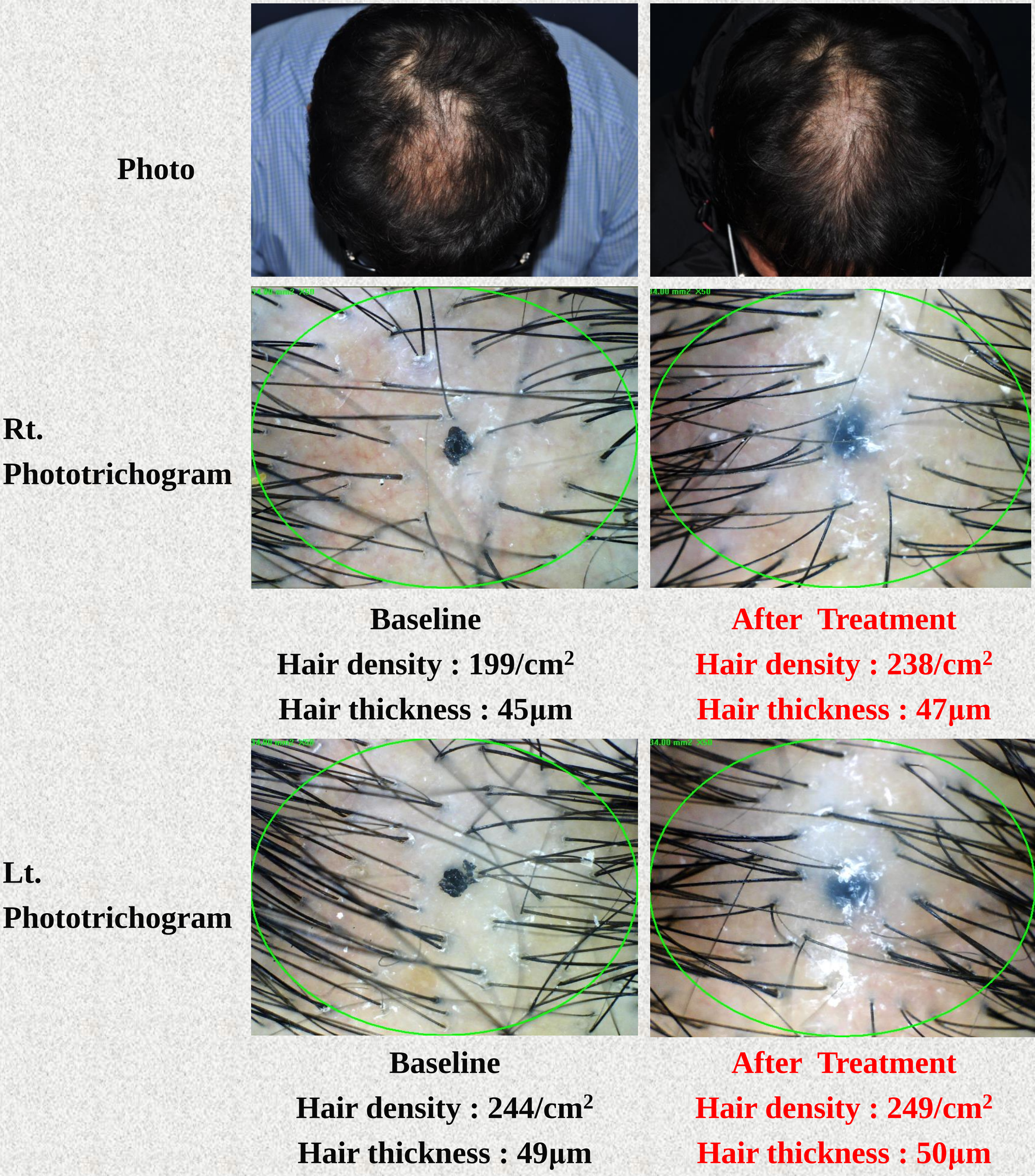
		Mean increase rate	p-value
Hair density	Rt.	13.8%	0.0033
	Lt.	8.9%	0.0038
Hair thickness	Rt.	4.8%	0.0032
	Lt.	3.3%	0.0033

Results

Differences of effects according to depth of microneedle

		P-value
Hair density	Rt. > Lt.	0.0352
Hair theickness	Rt. > Lt.	0.1117

M/46 MPHL III



Discussion

- Microneedle therapy is a recently presented therapeutic devise used for scalp treatment which increases topical treatment penetration and may independently promote new hair growth.
- Our study shows:
 - In density, 0.5mm depth was more effective than 0.3mm significant in difference($p<0.05$).
 - In thickness, effect according to depth was not significant in difference($p>0.05$).

Conclusion

- Our results show that 0.5 mm depth seems to be more effective than 0.3mm depth of microneedle.
- More research is necessary to prove result of this study.

References

- YB Lee, YB Eun, JH Lee et al. Effects of topical application of growth factors followed by microneedle therapy in women with female pattern hair loss: a pilot study. J Dermatol 39, 2012:21:231-4.
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